

Week 4 · Laboratory Solutions

- These are **reference** answers, not the only correct ones. The checker tests the physics, not the diagram.
- Read them after attempting the problems.

Building and checking

```
cd lectures/W04_simulink/solutions
W04_S1_heading_loop      % builds W04_S1.slx – all three problems in one model
W04_S_expected           % regenerates the expected-result figures

W04_check(1,'W04_S1')
W04_check(2,'W04_S1')
W04_check(3,'W04_S1')
```

All three pass and `check_overlaps('W04_S1')` is **0**. The output below is the measured one.

```
W04 problem 1
steady error at Kp = 100      0.0026 (expected  0.0000 +- 0.05 deg) PASS
overshoot at Kp = 100        5.7684 (expected  5.8000 +- 0.5 %) PASS
steady error at Kp = 300      0.0008 (expected  0.0000 +- 0.05 deg) PASS
overshoot at Kp = 300       12.1778 (expected 12.2000 +- 0.5 %) PASS

W04 problem 2
overshoot at Kd = 0           12.1778 (expected 12.1800 +- 0.5 %) PASS
overshoot at Kd = 50          4.1126 (expected  4.1000 +- 0.5 %) PASS
overshoot at Kd = 100         0.3929 (expected  0.3900 +- 0.5 %) PASS

W04 problem 3
turn WITH the wrap            19.9997 (expected 20.0000 +- 3 deg) PASS
turn WITHOUT the wrap        -339.9991 (expected -340.0000 +- 12 deg) PASS
```

The law, and the choices in it

$$e = \text{ssa}(\psi_d - \psi), \quad \tau_N = K_p e + K_i \int e \, dt + K_d \frac{N_f s}{s + N_f} e$$

	The choice	Why the alternative is worse
One model for three problems	the gains and <code>use_ssa</code> are variable names	problem 1 is $K_d = 0$, problem 2 is $K_d > 0$, problem 3 changes only <code>use_ssa</code> ; the checker sets them, so one model answers all three
The derivative is one Transfer Fcn	numerator <code>[Kd*Nf 0]</code> , denominator <code>[1 Nf]</code>	the same <code>D filter</code> as Weeks 2 and 3; a pure derivative would turn every sensor jitter and every step corner into a spike (Week 2 §2-10)
The error is wrapped before the gains	<code>ssa</code> in front of the Switch	without it a 20° command across the seam is executed as a 340° turn; problem 3 measures exactly that
The feedback is a Goto/From tag	<code>psi_fb</code>	a line crossing the whole model backwards lands on the forward path

Why the error is zero, and why D helps here

- The heading is the running sum of the turn rate: a constant yaw moment makes the heading grow for ever (lecture section C). The plant therefore already contains an integrator, and a proportional controller alone ends at the command. The speed of Week 3 had no such integrator, and P alone always left an error.
- The derivative of a heading error is a turn rate. Resisting it is damping, as on the mass of Week 2. The derivative of a speed error is an acceleration, and resisting that acts like extra mass — which is why the same term made Week 3 worse.

One measurement subtlety in problem 3

The checker reads ψ **unwrapped**, straight from state 12. A wrapped angle cannot tell $+20^\circ$ from -340° , because both end at the same physical heading; the quantity the problem is about is the **turn executed**. The two runs command from $t = 0$ so that they start with the same error and differ only in how that error is computed.